Applied Linear Algebra in Data Analysis

Positive Definite Matrices and Matrix Norms

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Positive Definite Matrices

There are matrices that are positive in one sense.

Positive definite matrices

We know that $\mathbf{x}^\top \mathbf{x} > 0$ for all $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{x} \neq \mathbf{0}$.

What can we say about $\mathbf{x}^\top \mathbf{A}\mathbf{x}$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$?

- Is it positive for all $\mathbf{x} \neq \mathbf{0}$? $\mathbf{I}$, $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$
- Can it be zero for some $\mathbf{x} \neq \mathbf{0}$? $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- Can it be negative some $\mathbf{x} \neq \mathbf{0}$? $\begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$, $\begin{bmatrix} 1 & -4 \\ -4 & 1 \end{bmatrix}$

Any matrix $\mathbf{A}$ for which $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$, for all $\mathbf{x} \neq \mathbf{0}$ is called a *positive definite matrix*.

Positive definite matrices are commonly encountered in practice: optimization, mechanics (mass matrix, stiffness matrix), stability analysis, co-variance matrices etc.

Positive definite matrices

The idea of positive definiteness is intimately related to the problem of minimization of a function.

Consider the following function of a single variable $f(x)$. This function reaches a minimum at $x = 0$, when $\frac{df(x)}{dx}\Big|_{x=0} = 0$ and $\frac{d^2f(x)}{dx^2}\Big|_{x=0} > 0$. E.g.,

$$f(x) = 3x^2 \longrightarrow \frac{df(x)}{dx}\Big|_{x=0} = 0, \ \frac{d^2f(x)}{dx^2}\Big|_{x=0} = 3 > 0$$

Positive definite matrices

What about $f(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2$? We can extend the previous idea using partial derivatives.

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 0, \quad \frac{\partial f(x_1, x_2)}{\partial x_2} = 0, \quad \frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} > 0, \quad \frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} > 0$$

This is not enough. $\frac{\partial^2 f(x_1,x_2)}{\partial x_1 \partial x_2}$ must also be taken into account.

Are these functions positive for all x_1, x_2?

1. $x_1^2 + x_1x_2 + x_2^2$
2. $x_1^2 + 2x_1x_2 + x_2^2$
3. $x_1^2 + 3x_1x_2 + x_2^2$?

Positive definite matrices

We can rearrange $ax_1^2 + 2bx_1x_2 + cx_2^2$ in the following manner,

$$f(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2 = a\left(x_1 + \frac{b}{a}x_2\right)^2 + \left(c - \frac{b^2}{a}\right)x_2^2$$

$f(\bullet) > 0, \forall x_1, x_2 \neq 0$ when,

$$a > 0 \quad \text{and} \quad c - \frac{b^2}{a} > 0 \implies ac > b^2$$

$$\frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} > 0 \quad \text{and} \quad \frac{\partial^2 f(x_1, x_2)}{\partial x_1^2}\frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} > \left(\frac{\partial^2 f(x_1, x_2)}{\partial x_1 \partial x_2}\right)^2$$

Positive definite matrices

$f(\bullet)$ can be expressed as $\mathbf{x}^\top \mathbf{A}\mathbf{x}$, where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $\mathbf{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$. $f(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$, when $\mathbf{A}$ is positive definite.

$\mathbf{x}^\top \mathbf{A}\mathbf{x}$ is called a *quadratic form*. For a symmetric matrix $\mathbf{A}$,

$$\mathbf{x}^\top \mathbf{A}\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} = \sum_{i=1}^{n} a_{ii} x_i^2 + 2 \sum_{i=1}^{n} \sum_{j=1, j \neq i}^{n} a_{ij} x_i x_j$$

Positive definite matrices

In general, $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive definite, if:

- The eigenvalues of $\mathbf{A}$ are all positive.
- The pivots (without row exchange) are all positive.

Show that any $\mathbf{A}$ is positive definite if the symmetric matrix $\mathbf{A} + \mathbf{A}^\top$ is positive definite.
Note: *This should explain why we have only been talking about symmetric matrices.*

Matrix Norms

How big is a matrix?

Matrix Norm

Since the set matrices $\mathbf{A} \in \mathbb{R}^{p \times q}$ form vector spaces, we can talk about norms of matrices, extending the idea of sizes and distances to spaces of matrices.

If we think of matrices a collection of pq scalars, then we can use the same approach as vectors,

$$\|\mathbf{A}\|_F = \left(\sum_{i=1}^{p} \sum_{j=1}^{q} |a_{ij}|^2 \right)^{\frac{1}{2}}$$

This is called the **Frobinius norm**.

Note that the Frobinius norm is the 2-norm equivalent.

Induced Matrix Norm

There are other norms defined for matrices that are very useful from the point of the view of linear transformation.

These are called **induced matrix norms**, that looks at how matrices map vectors from the range to domain spaces.

Let $\mathbf{A} \in \mathbb{R}^{p \times q} : \mathbb{R}^q \longrightarrow \mathbb{R}^p$, and $\mathbf{x} \in \mathbb{R}^q$, $\mathbf{y} \in \mathbb{R}^p$

$$\|\mathbf{y}\| = \|\mathbf{A}\mathbf{x}\| \leq C \|\mathbf{x}\|, \quad \forall \mathbf{x} \in \mathbb{R}^q, \quad C \geq 0$$

C is the maximum factor by which $\mathbf{A}$ amplifies the vector $\mathbf{x}$.

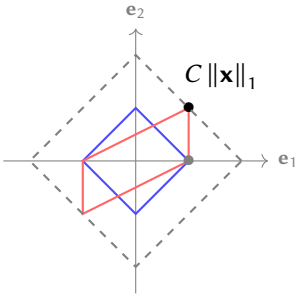
The induced norm of a matrix is defined as,

$$\|\mathbf{A}\|_p = \max_{\mathbf{x}} \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p} = \max_{\|\mathbf{x}\|_p = 1} \|\mathbf{A}\mathbf{x}\|_p$$

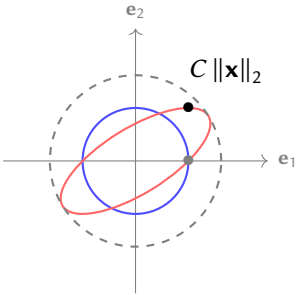
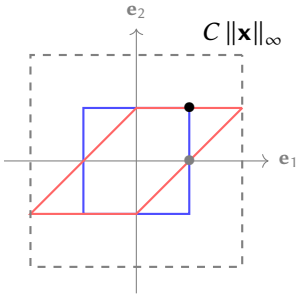
Induced Matrix Norm

Consider a matrix $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{a}}_1^\top \\ \tilde{\mathbf{a}}_2^\top \end{bmatrix}$.

$\|\mathbf{A}\|_1 = \max_i \|\mathbf{a}_i\|_1$



$\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}}$



$\|\mathbf{A}\|_\infty = \max_i \|\tilde{\mathbf{a}}_i\|_1$